SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	29th June 2026
Team ID	xxxxxx
Project Name	Rising waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/238x1a05b7/Rising-waters
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost

Key Features Implemented	Flood risk prediction using Machine Learning. • User-friendly web interface for entering environmental parameters. • Real-time prediction with probability scores.
Pending / Incomplete Features	Real-time weather API integration. • GPS/location-based flood prediction. • SMS/Email alert notifications.
Setup / Run Instructions	1. Clone the repository from GitHub. 2. Install the required packages using pip install -r requirements.txt. 3. Run the application using python app.py. 4. Open the local URL (e.g., http://127.0.0.1:5000) in a web browser. 5. Enter the environmental parameters and click **Predict** to view the flood risk prediction.

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: